RX2309(12A)-COMP Specification

Ver. A

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Change Summary：

SN	Original version number	Date of change	Changed Content	Reason for change	Changed by	Changed version
1	A	2023.06.18			Hu Lijun	A

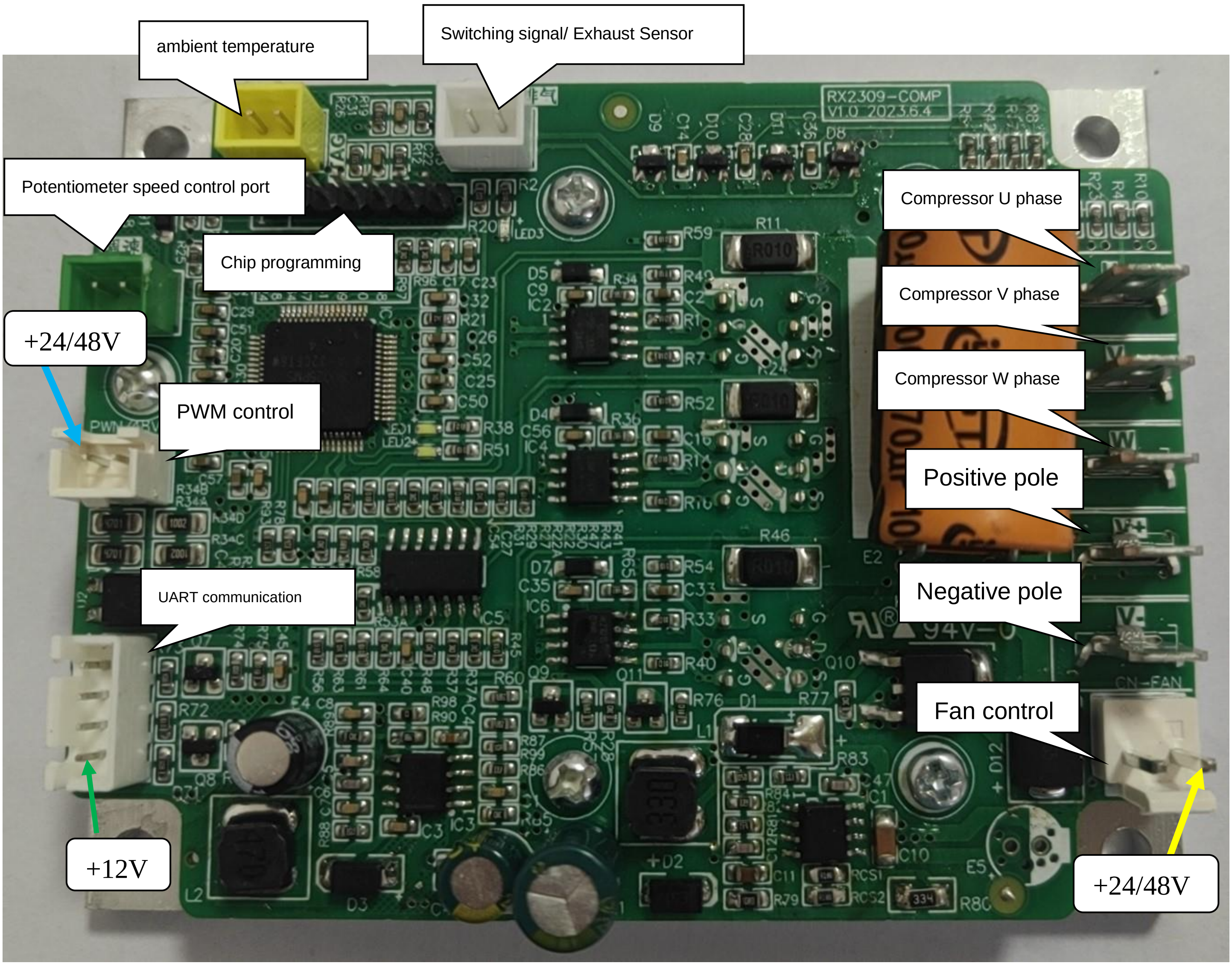
RX2309-COMP compressor controller adopts TI DSP 280025 as the driver chip and FOC (Field Oriented Control) 180° control method. Minimum operating speed reaches 60 rpm (under high load conditions). Sampling high modulation rate is adopted, at high speed compared with the industry energy efficiency ratio increased by about 5%.

1.Technical Requirements

Project	规格	
Applicable object	24V/48V compressor	
Max. input current	12A	
Max. input power	24V*12A or 48V*12A	
Min. speed	1800rpm (different compressor minimum speed is different)	
Max. speed	6500rpm (different compressor highest speed is different)	
Compressor speed accuracy	±10rpm	
Rated input voltage	24VDC/48VDC (Maximum device input voltage is 100V)	
Power input range	24VDC	18VDC~32 VDC (Battery protection voltage) Software can be adapted to user requirements.
	48VDC	41VDC~65 VDC (Battery protection voltage), Software can be adapted to user requirements.
Frequency conversion efficiency	>90%	
Cooling mode	Heat sink cooling	
Operating environment	Temperature: -20℃ ~ 55℃ Humidity: 30% ~ 90%	
Protective function	Software overcurrent protection: compressor controller output U\V\W any phase greater than 28A (peak) overcurrent protection;	
	Over-voltage protection: input voltage greater than 32V, over-voltage protection; (24V working voltage) Over-voltage protection recovery voltage: input voltage less than 30V; recovery over-voltage protection; (24V working voltage)	
	Over-voltage protection: input voltage greater than 65V, over-voltage protection; (68V working voltage) Over-voltage protection recovery voltage: input voltage less than 64V; recovery over-voltage protection; (48V working voltage)	
	Undervoltage protection: input voltage less than 18.0V, low voltage protection; (24V working voltage) Undervoltage protection recovery voltage: input voltage higher than 23V; recovery low voltage protection; (24V working voltage)	
	Undervoltage protection: input voltage less than 41V, low voltage protection; (48V working voltage) Undervoltage protection recovery voltage: input voltage higher than 45V; recovery low voltage protection; (48V working voltage)	
	Out-of-phase protection: When compressor is driving, out-of-phase protection is adopted if detecting the positioning current U\V\W any phase current less than 2A;	
	Hardware overcurrent protection: Overcurrent protection is adopted if compressor controller output U\V\W any one phase is greater than 40A	
	Compressor stall or startup failure protection: When the calculated speed of the compressor is greater than 8000 RPM, or	

	less than 0 RPM for 5 seconds to report stall protection. When the compressor driver board internal setting target speed and feedback speed difference is 200 rpm for 20 seconds, reported stall protection;
	Phase current anomaly protection: compressor (U\V\W) phase current is 2 seconds less than 0.4A or for 2 seconds more than 5A, reported compressor phase current output anomalous
	Tilt protection: controller tilt switch not installed, no
Protective action	Over-voltage and under-voltage protection function, when the voltage is higher or lower than a certain value, the controller does not work, in a standby state.
	Current protection function, when the current is higher than a certain value, the controller does not work, in a standby state.
	When the controller starts, the compressor three-phase detection of a phase or multi-phase is not linked well, the controller does not work, in a standby state.
	When the controller starts, check the compressor start failure or stall state, the controller does not work, in a standby state.
	When the controller starts up and detects abnormal phase currents, the controller does not work, in standby mode.
	When the controller detects protection, stop running and restart after 1 minutes. (Voltage protection excepted)
Driving mode	Forward wave drive
Interface mode	UART communications (Reserved)
Control panel dimensions	96*57 (mm)
Heat sink dimensions	100*77 (mm)

2.Description of controller connection ports



Interface description:

1.UART communication: used for communication between the compressor communication board and the operation board.

2.Compressor resistance commissioning: (reserved): Use 51KΩ for regulating the compressor running speed.

3.Switching signals: used for potentiometer speed control, disconnect shuts off the compressor, short circuit enables compression

4.Temperature sensor: The discharge temperature sensor detects the compressor discharge temperature. (Reserved)

5.Chip programming: simulation and program curing of the chip.

6.Positive pole: Connect the Positive pole.

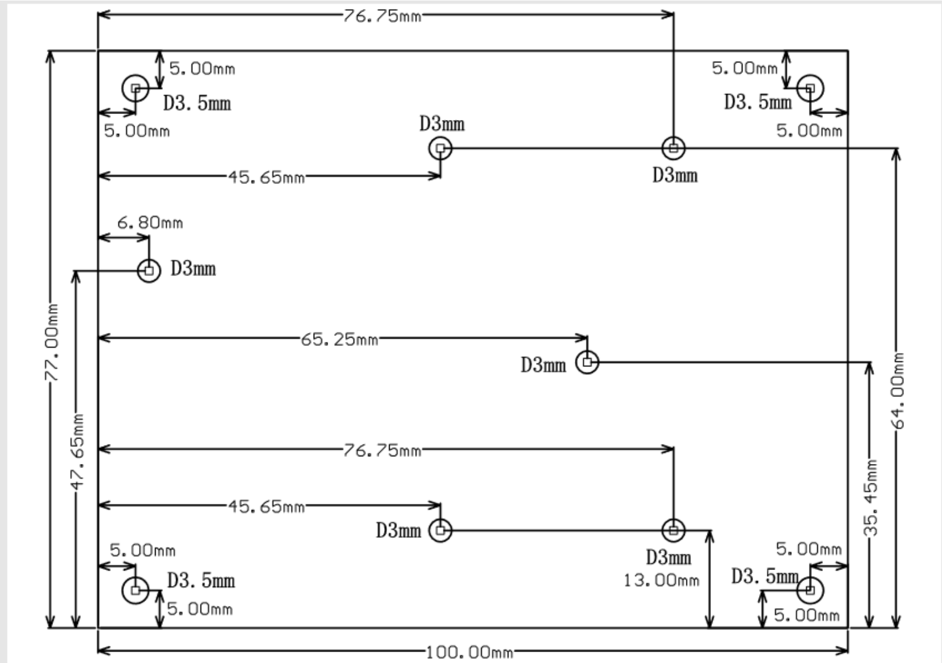
7.Negative pole: Connect the Negative pole.

8.Compressor output port

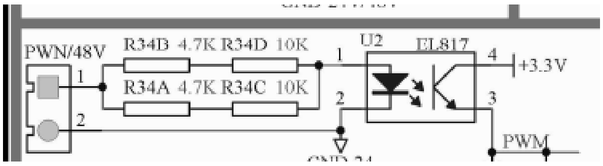
Compressor U phase, compressor V phase, compressor W phase: connect compressor U, V, and W phases.

9.Fan Control: Controls the DC fan on and off.

3.Heat sink size drawing



PWM drawing



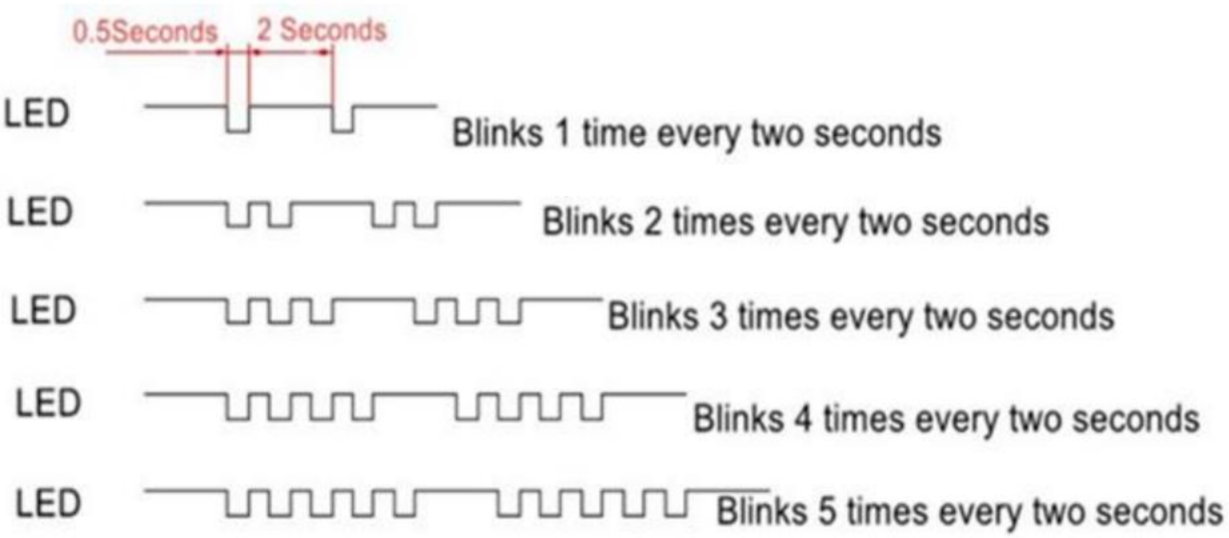
Pulse control frequency. The control frequency corresponds to the compressor speed

Control frequency (HZ)	Compressor speed (RPS)	Compressor (RPM)
56	28	1680
80	40	2400
100	50	3000
120	60	3600
140	70	4200
160	80	4800
180	90	5400
200	100	6000

Input pulse signal for control is 50% square wave signal, column as 50HZ control frequency, input pulse signal high level 10ms, low level 10ms. target speed (RPS) = 25*60 = 1500 (RPM).

The fault indicator (LED1) flashes with the feedback pulse with down:

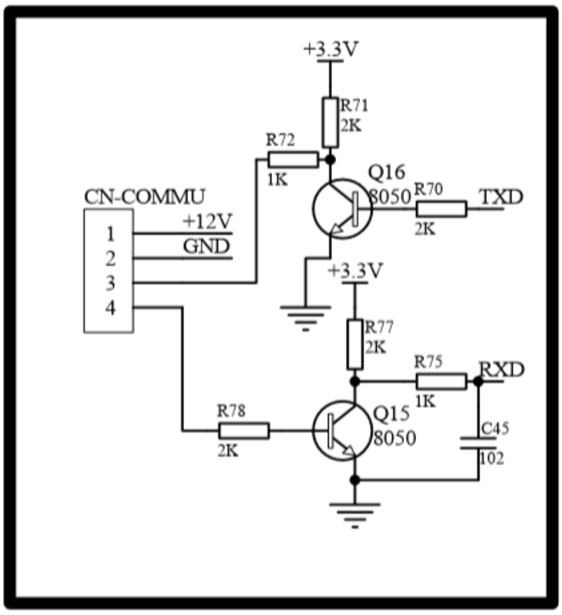
LED2: Blinks when the compressor is running, stops when the compressor is stopped;



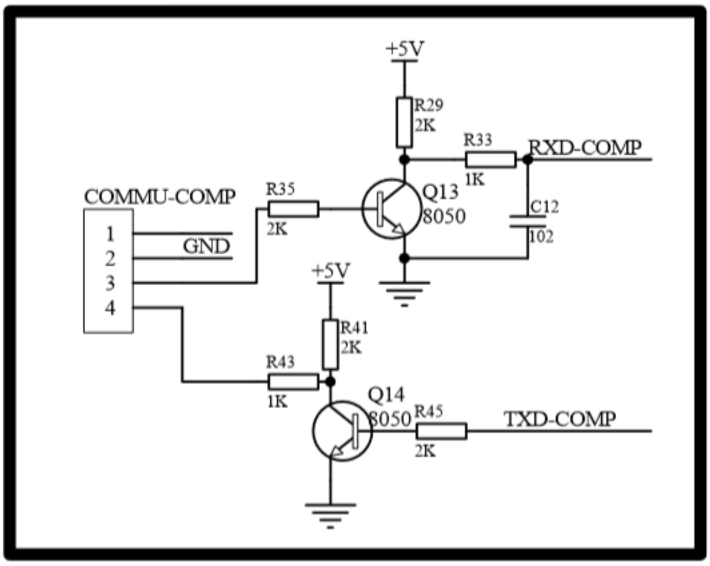
Blink 1time: Compressor stall or Compressor blockage
Blink 2 times: Lack-phase
Blink 3 times:Over current protection
Blink 4 times: Low Voltage or Over Voltage
Blink 5 times: MOS over-temperature protection

UART communication：

The driver board UART communication circuit is as follows:



The reference circuit of the upper computer is as follows: (User upper computer circuit)



Communication Protocols

The communication between the driver board and the Main Control Board adopts the masterslave communication mode. The main control is the initiator of the communication, and the communication adopts UART mode. The main machine is the Operation Board, and the slavemachine is the compressor driving board.

Baud rate: 600bp

Data format: 1 bit starts bit, 8 bits data, 1 bit stops

The master issued 16 bytes peframe, 16 bytes from the machine reply. The master emits a frame every 1000ms, and receives a frame from the master with a 20ms delay in response to a frame. The master address is 0x00 and the slave address is 0x01.

0	0xAA	Start code
1	0X00	
2	Instructions	Bit0: 1:on, 0:off;
3	Set speed	Low byte
4	Set speed	High byte
5	Reserve	
6	Reserve	
7	0x00	
8	0x00	
9	0x00	
10	0x00	
11	0x00	
12	0x00	
13	0x00	
14	Checksum	(byte1+byte2+……byte13) reverse +1
15	0x55	End code

Example:

The set speed is set to 3000 rpm and the data is sent as follows with the compressor on:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0xAA	0x00	0x01	0xB8	0x0B	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x3c	0x55

The set speed is set to 0 rpm and the off compressor sends data as follows:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0xAA	0x00	0x00	0x0	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x55

Data returned from the slave

0	0xAA	Start code
1	0X01	
2	Compressor rotation speed	Low byte
3	Compressor rotation speed	High byte
4	Compressor current	Low byte，The precision of 0.1A
5	Compressor current	High byte
6	busbar voltage	Low byte，The precision of 0.1V
7	busbar voltage	High byte
8		
9	error code (This faulty unit is not automatically cleared, and the compressor starts to clear)	Bit0: software overcurrent Bit1: Overvoltage protection Bit2: Undervoltage protection Bit3: Phase loss protection Bit4: Stall protection Bit5: Hardware overcurrent protection Bit6: Abnormal phase current Bit7:
10	Ambient temperature	Reserve
11	Ventilate	Reserve
12	0x00	
13	0x00 The upper computer reads the faulty unit, and the faulty drive board of this unit is automatically cleared in 120 seconds.	Bit0: software overcurrent Bit1: Overvoltage protection Bit2: Undervoltage protection Bit3: Phase loss protection Bit4: Stall protection Bit5: Hardware overcurrent protection Bit6: Abnormal phase current Bit7:
14	Checksum	(byte1+byte2+……byte13) reverse +1
15	0x55	End code

Compressor command is on if the host sets speed 3000 rpm.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0xAA	0x00	0x01	0xB8	0x0B	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x3c	0x55

Driver board compressor starts response data: e.g. actual compressor speed is 2998 rpm, input voltage 26.1V, input current 4A

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0xAA	0x01	0xB6	0x0B	0x28	0x00	0x05	0x01	0x00	0x00	0x00	0x00	0x00	0x00	0x10	0x55